CMAI 480

Computer Vision

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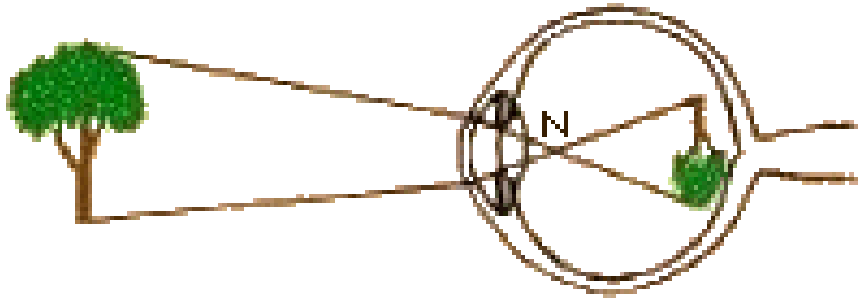
Outline

- Geometric primitives and transformations
  - Review of Linear Algebra
  - Projection geometry
    - Homogeneous Representation of 2D/3D entities or features
    - 2D/3D Image Transformations
- Human vision system
- Human vision for computer vision
- Cameras and image formation

HUMAN VISUAL SYSTEM

Human Eyes

- The human eye is the organ which gives us the sense of sight, allowing us to observe and learn more about the surrounding world than we do with any of the other four senses.
- The eye allows us to see and interpret the shapes, colors, and dimensions of objects in the world by processing the light they reflect or emit. The eye is able to detect bright light or dim light, but it cannot sense objects when light is absent.

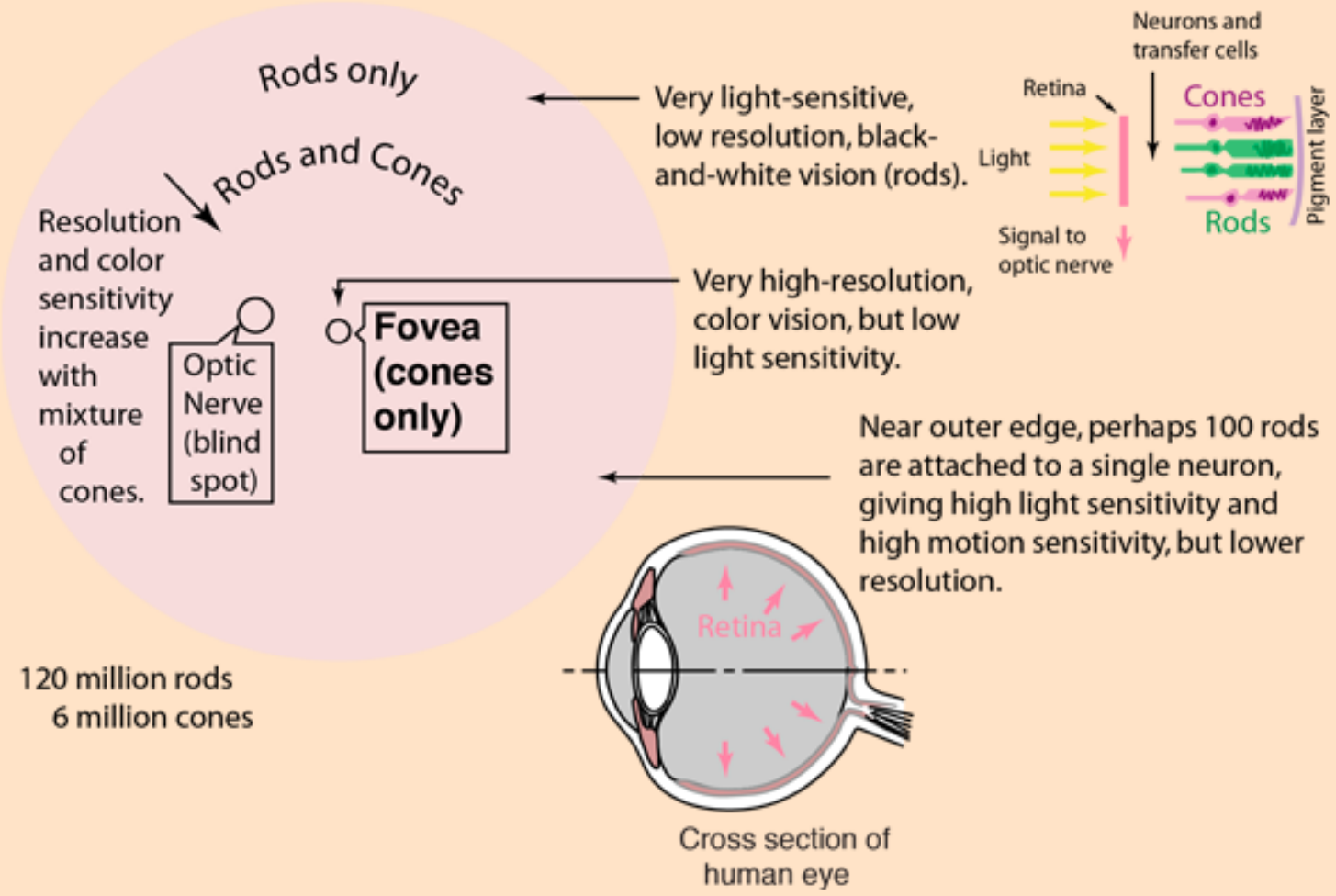


Human Eyes

- Retina – The retina is the innermost layer of the eye and is comparable to the film/camera sensor inside of a camera.
  - It is composed of nerve tissue which senses the light entering the eye.

The Retina

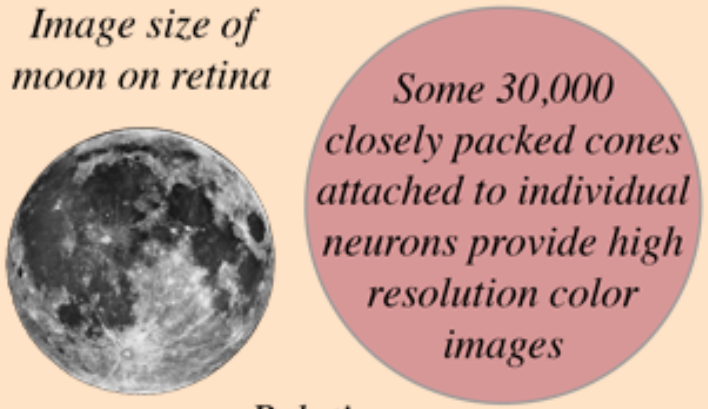
The retina is a light-sensitive layer at the back of the eye that covers about 65 percent of its interior surface. Photosensitive cells called rods and cones in the retina convert incident light energy into signals that are carried to the brain by the optic nerve. In the middle of the retina is a small dimple called the fovea or fovea centralis. It is the center of the eye's sharpest vision and the location of most color perception.

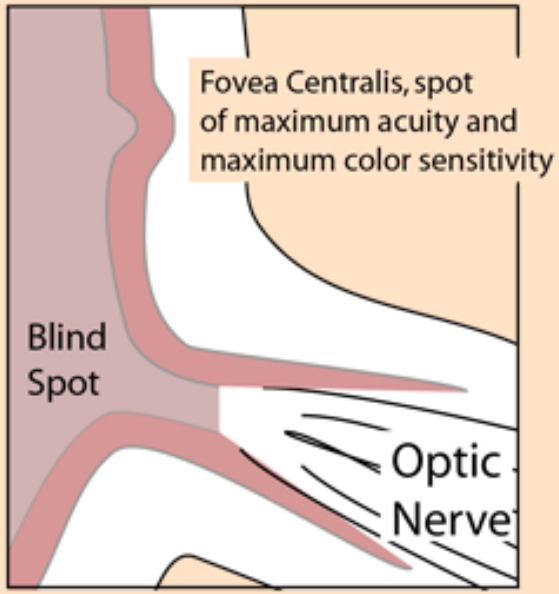


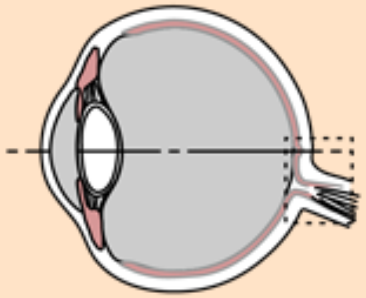
Human Eyes

The Fovea Centralis

The eye's sharpest and most brilliantly colored vision occurs when light is focused on the tiny dimple on the retina called the fovea centralis or macula. This region has exclusively cones and they are smaller and more closely packed than elsewhere on the retina.

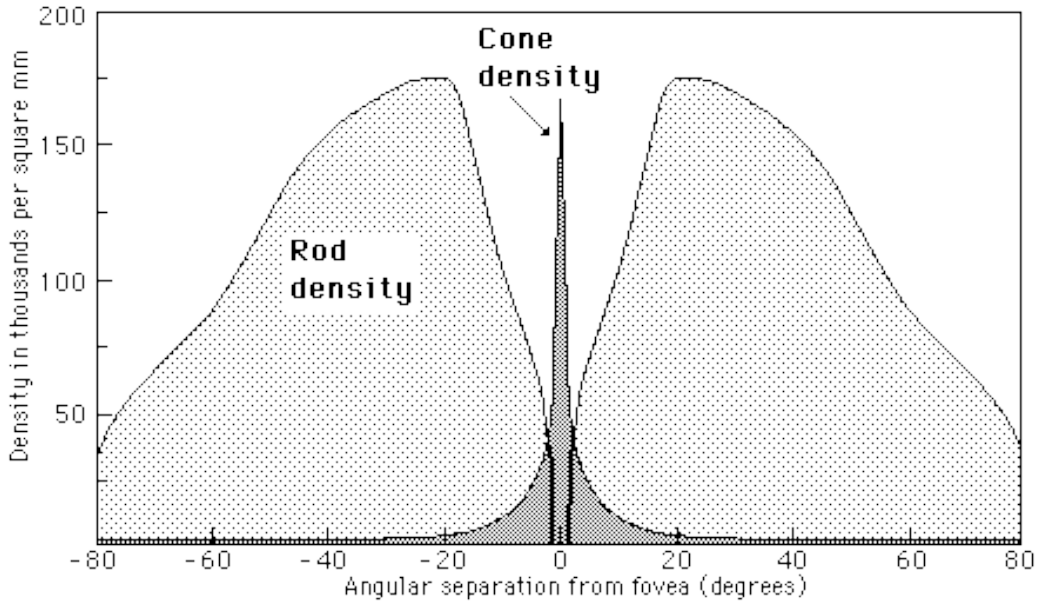





Cross section of human eye

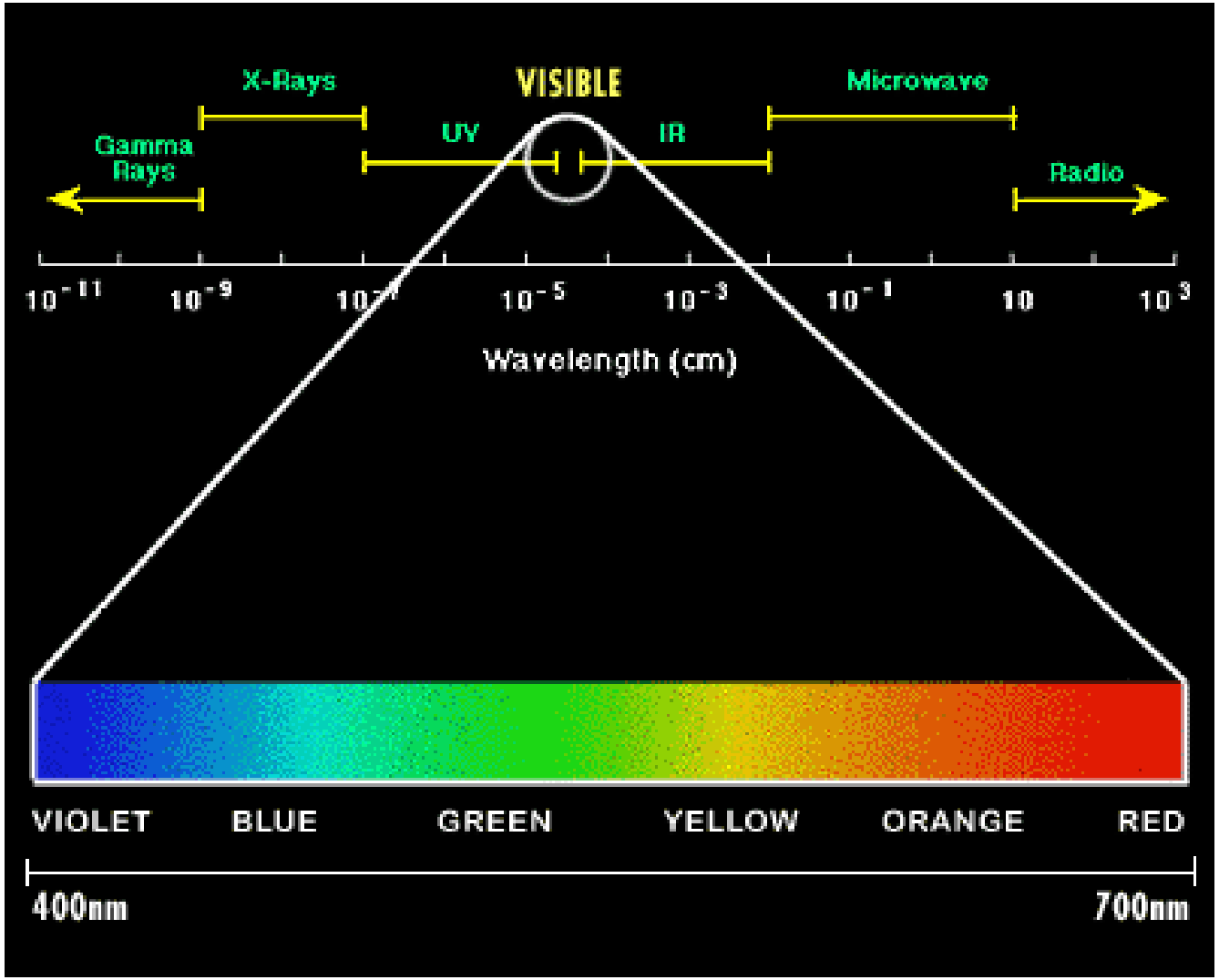
Rods and Cones

- The retina contains two types of photoreceptors, rods and cones.
- The rods are more numerous, some 120 million, and are more sensitive than the cones. However, they are not sensitive to color.
- The 6 to 7 million cones provide the eye's color sensitivity and they are much more concentrated in the central yellow spot known as the macula.



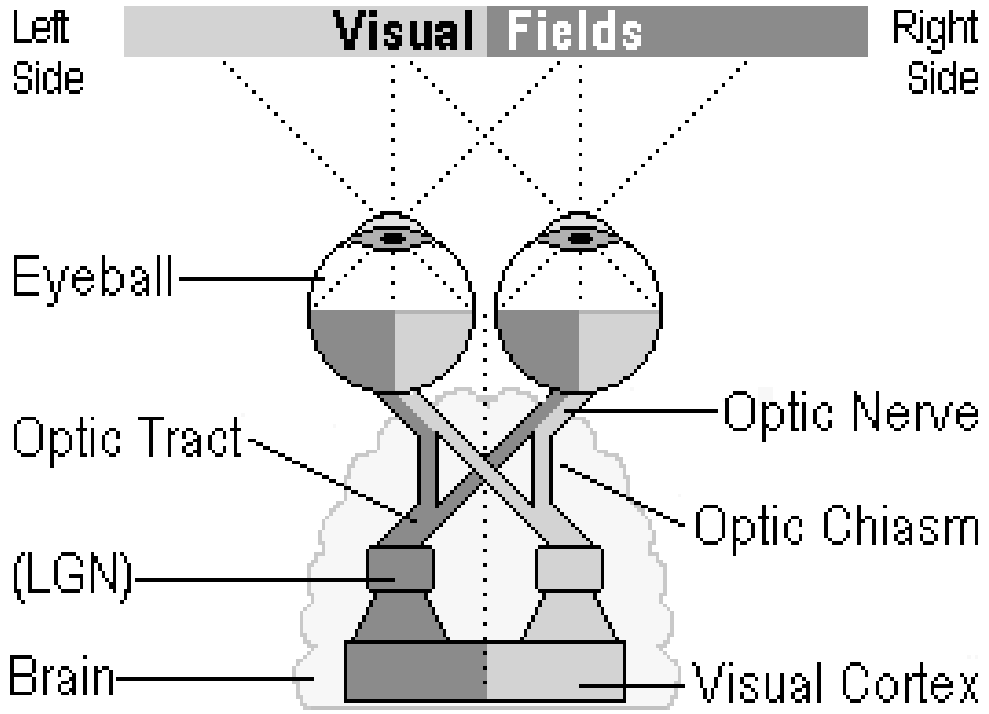
Photoreceptors Sensitivity to Light

The Electromagnetic Spectrum (measured by Wavelength)



Human Vision System

- We do not “see” with our eyes, but with our brains



Human Vision for Computer Vision

"Human vision is vastly better at recognition than any of our current computer systems, so any hints of how to proceed from biology are likely to be very useful."

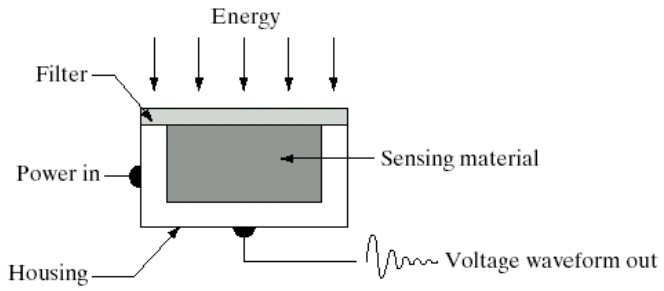
– David Lowe

Image Sensors

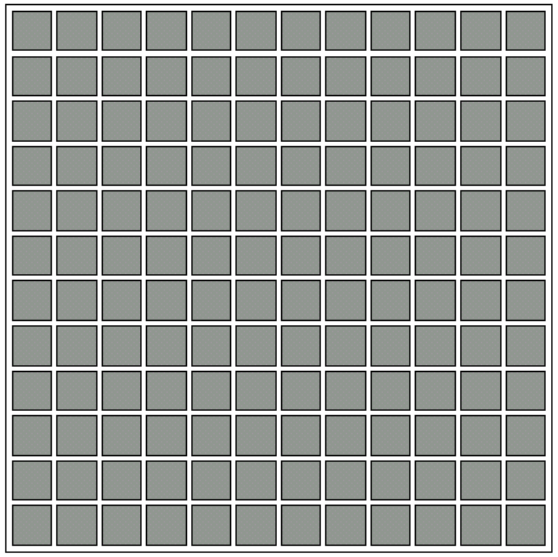
- Incoming energy (reflected or transmitted through) is transformed into a voltage.
- Sensor material is responsive to a particular type of energy being detected
- Sensor is organized into cells or true pixels, which defines the resolution of the camera

a
b
c

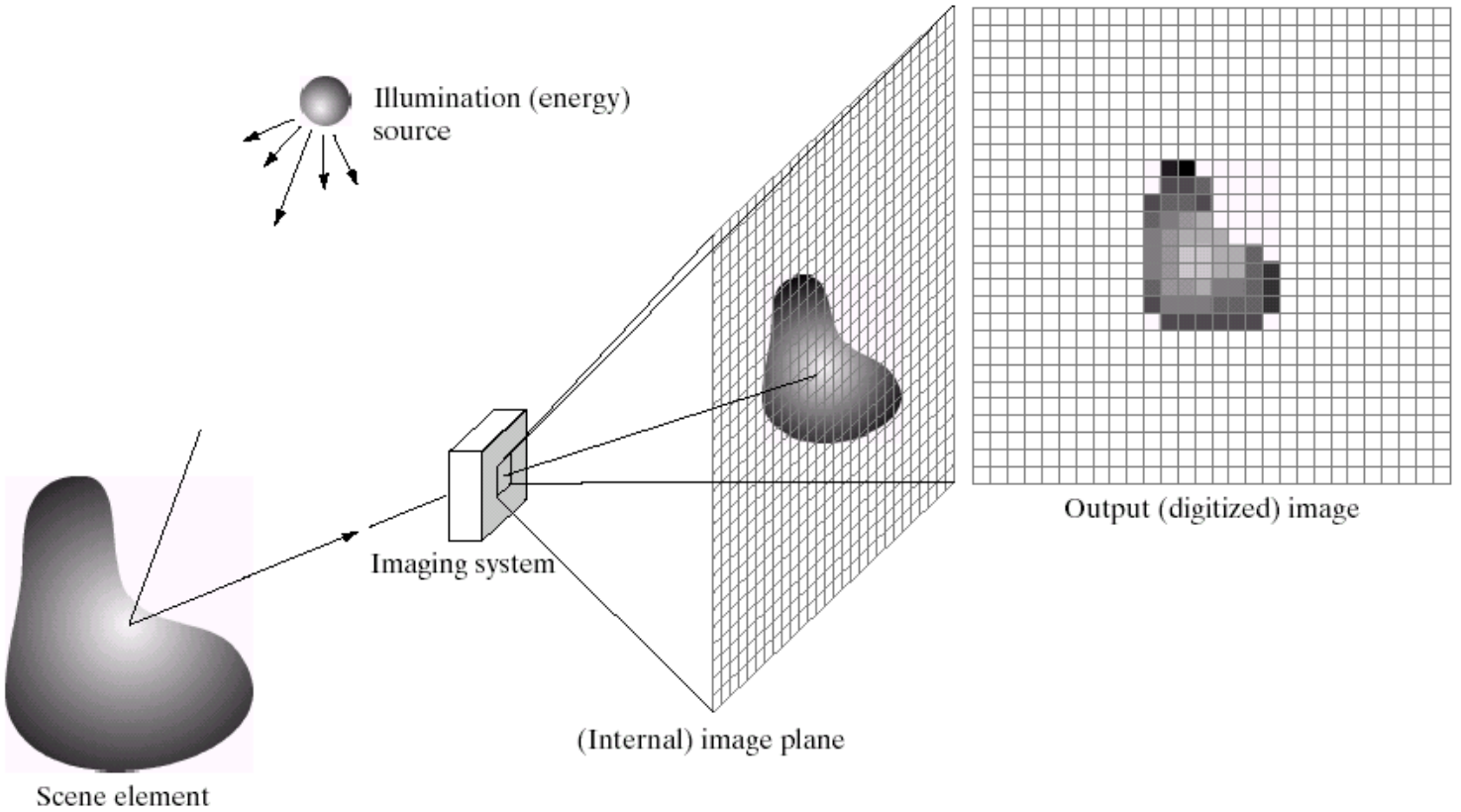
FIGURE 2.12
(a) Single imaging sensor.
(b) Line sensor.
(c) Array sensor.







An Example of Digital Image Acquisition Process: Sensors Array

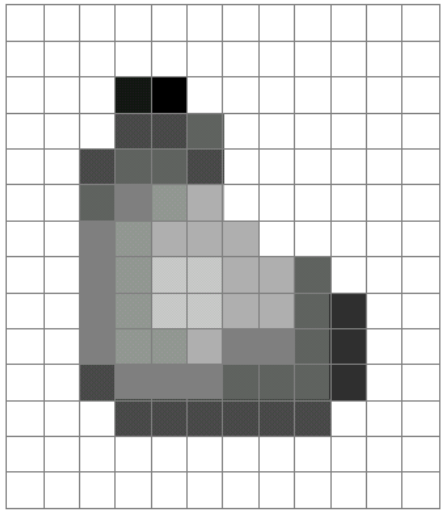


a b c d e

FIGURE 2.15 An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

What is an image?

- A grid (matrix) of intensity values



=

255	255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255	255
255	255	255	20	0	255	255	255	255	255	255	255
255	255	255	75	75	75	255	255	255	255	255	255
255	255	75	95	95	75	255	255	255	255	255	255
255	255	96	127	145	175	255	255	255	255	255	255
255	255	127	145	175	175	175	255	255	255	255	255
255	255	127	145	200	200	175	175	95	255	255	255
255	255	127	145	200	200	175	175	95	47	255	255
255	255	127	145	145	175	127	127	95	47	255	255
255	255	74	127	127	127	95	95	95	47	255	255
255	255	255	74	74	74	74	74	74	255	255	255
255	255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255	255

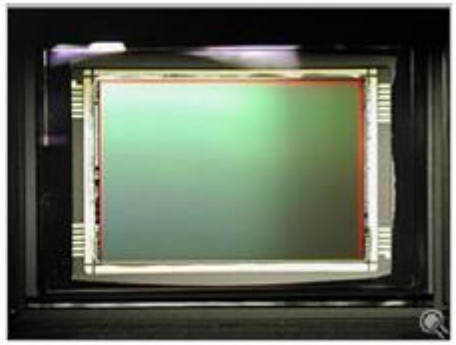
(common to use one byte per value: 0 = black, 255 = white)

Digital Camera

More details are in upcoming classes

- A camera is a mapping from the 3D world to a 2D image.





This is very similar to our Human Visual System

CCD Vs. CMOS camera Sensors

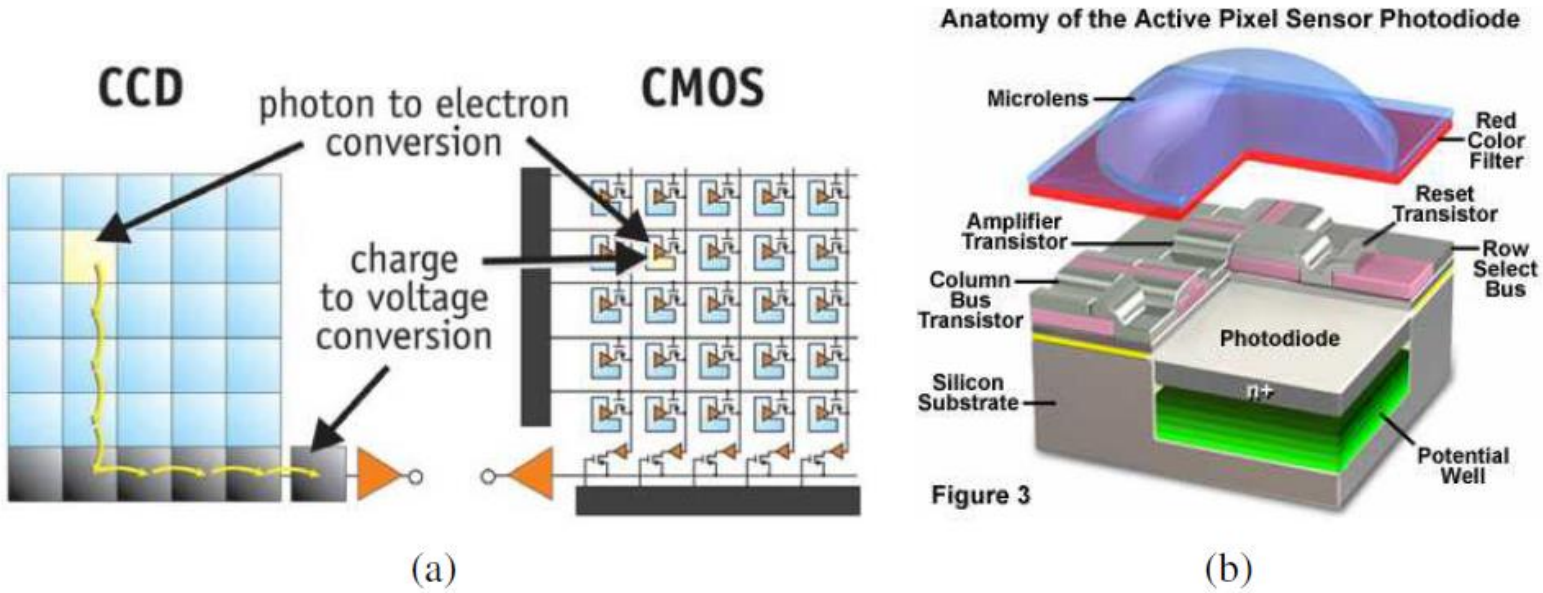


Figure 2.24 *Digital imaging sensors: (a) CCDs move photogenerated charge from pixel to pixel and convert it to voltage at the output node; CMOS imagers convert charge to voltage inside each pixel (Litwiller 2005) © 2005 Photonics Spectra; (b) cutaway diagram of a CMOS pixel sensor, from https://micro.magnet.fsu.edu/primer/digitalimaging/cmosimagesensors.html.*

Image Sensing Pipeline

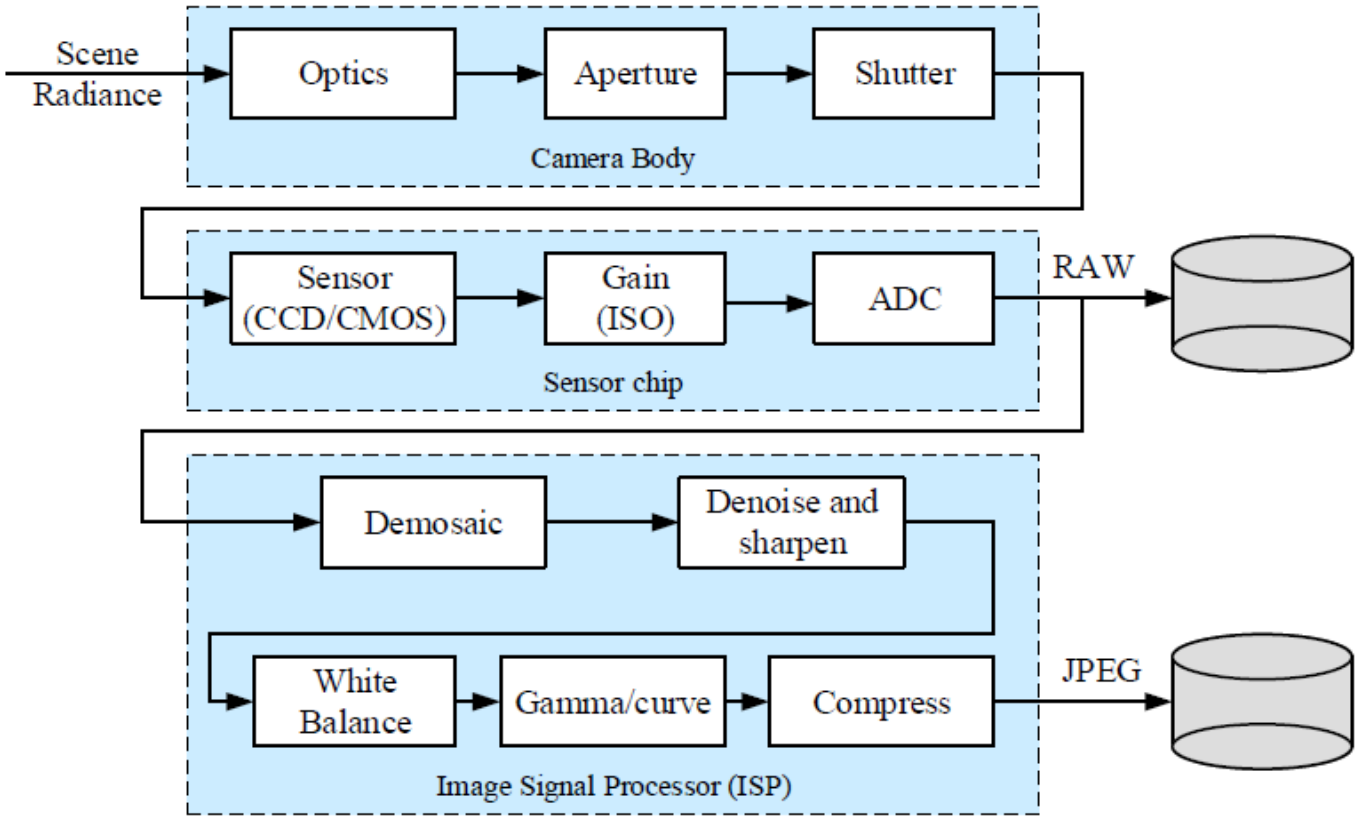


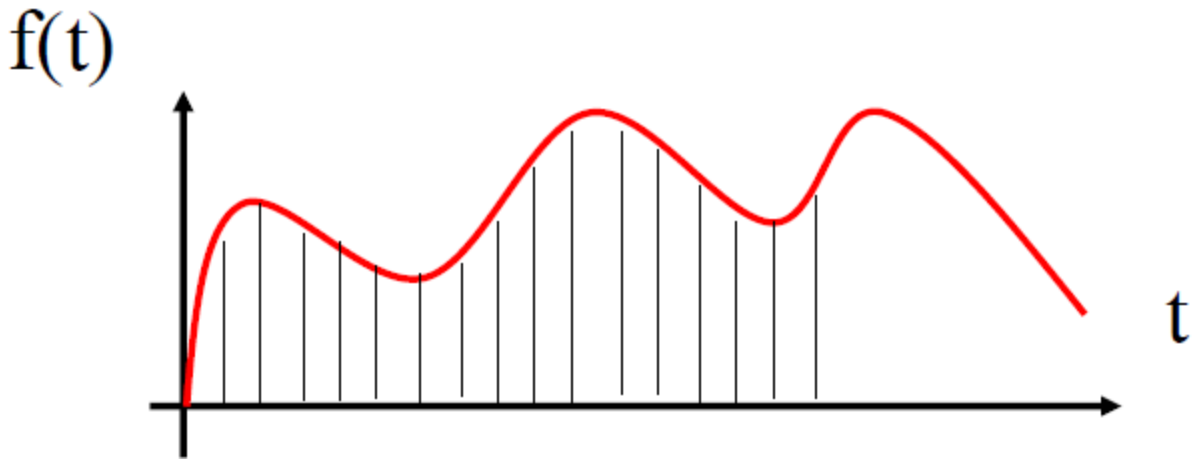
Figure 2.23 *Image sensing pipeline, showing the various sources of noise as well as typical digital post-processing steps.*

Digital Image Generation

- Two Main Steps:
  - Sampling and
  - Quantization

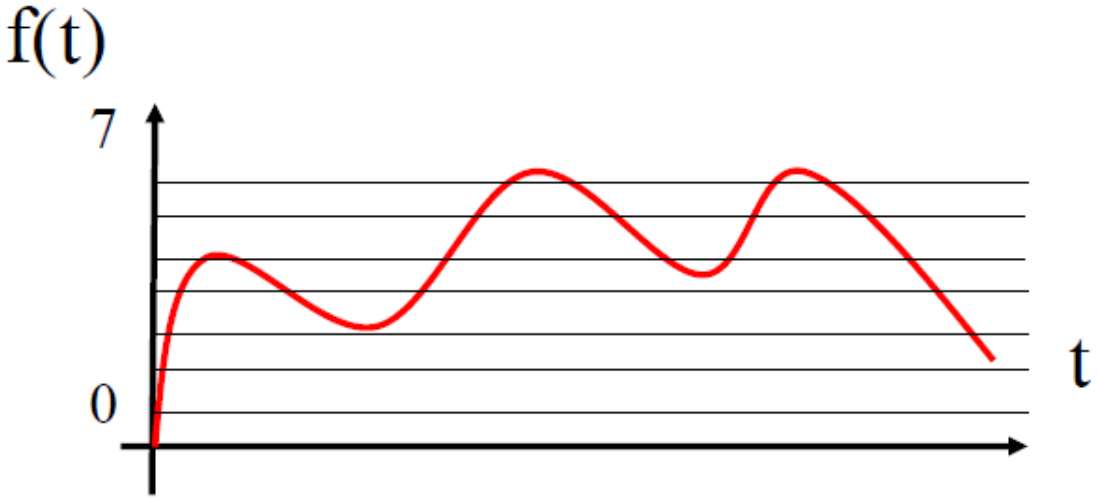
Sampling

- Sampling = the spacing of discrete values in the domain of a signal.
- Sampling-rate = how many samples are taken per unit of each dimension. e.g., samples per second, frames per second, etc.



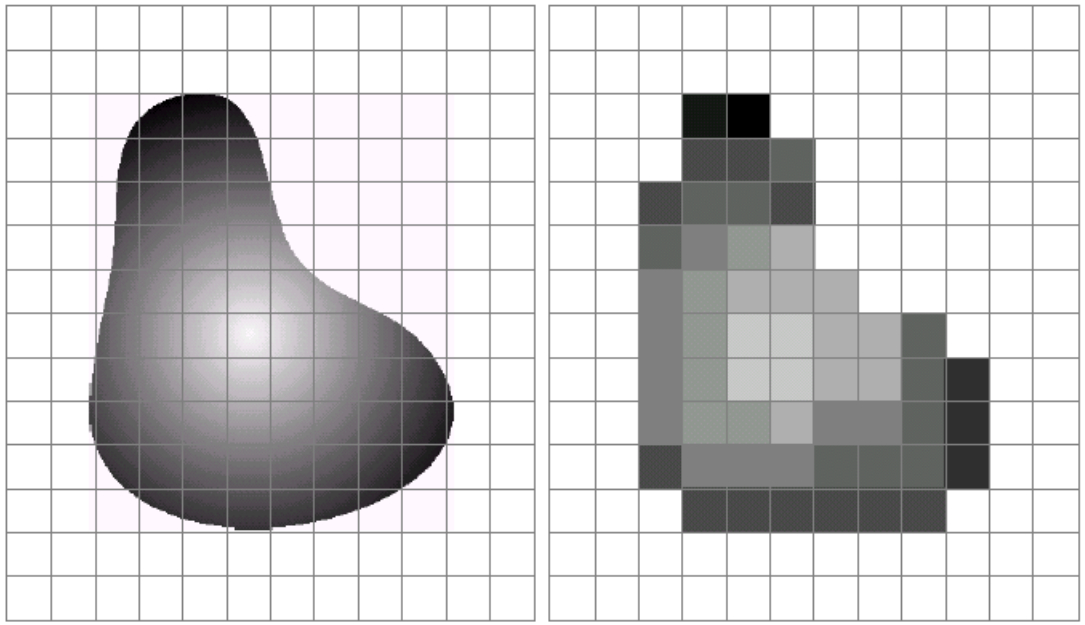
Quantization

- Quantization = spacing of discrete values in the range of a signal.
- Usually thought of as the number of bits per sample of the signal. e.g., 1 bit per pixel (b/w images), 16-bit audio, 24-bit color images, etc.



8 levels = 23 : uses 3 bits to represent the values of the function.

Image Sampling and Quantization



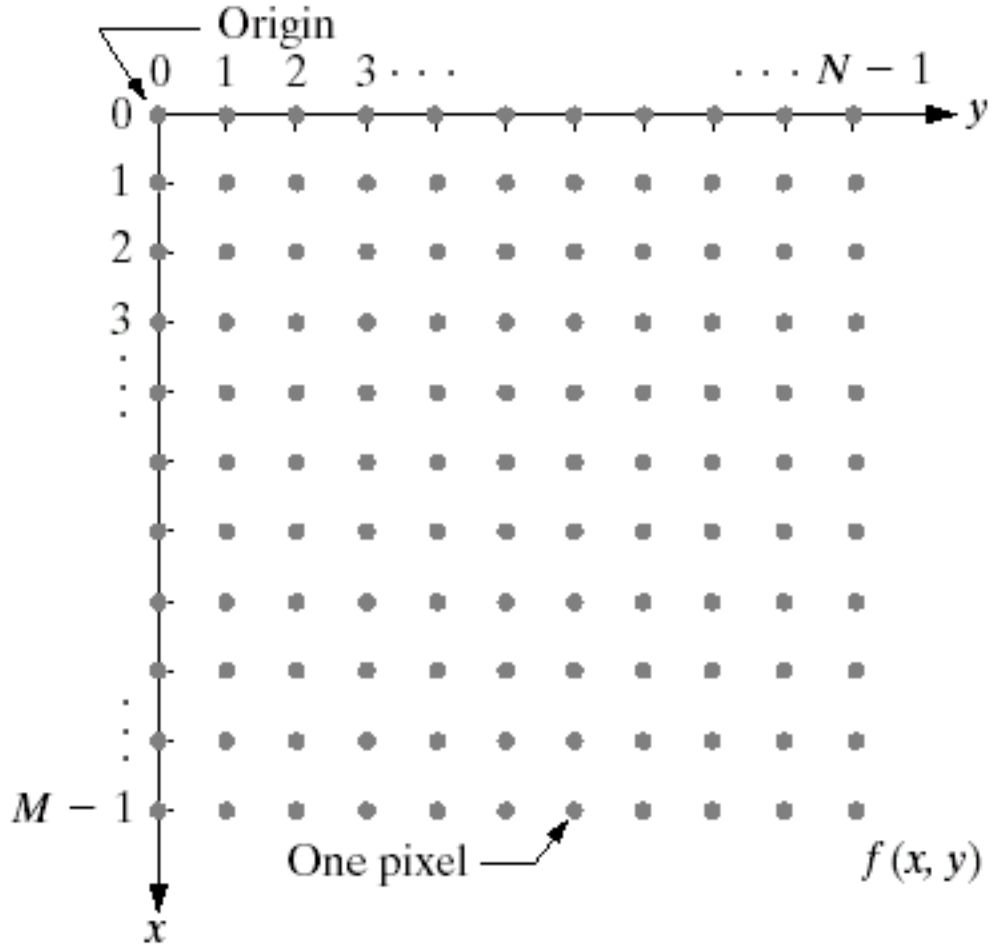
a b

FIGURE 2.17 (a) Continuos image projected onto a sensor array. (b) Result of image sampling and quantization.

Quality depends on:
- Number of samples.
- Number of gray levels.

Digital Image Representation

- An image is a function defined on a 2D coordinate $f(x,y)$.
- The value of $f(x,y)$ is the intensity.
- 3 such functions can be defined for a color image, each represents one color component
- A digital image can be represented as a matrix (MxN).



Matlab Exercise

- Use Matlab to read and show an image (Can also use OpenCV)
  - imread()
  - imshow()

First Measure: Quantization
Number of Gray Levels

- Can be viewed as the range that represents black and white
  - i.e. how many levels is there between black and white
  - Original: 5 levels
  - 16 levels
  - Now: 256 levels

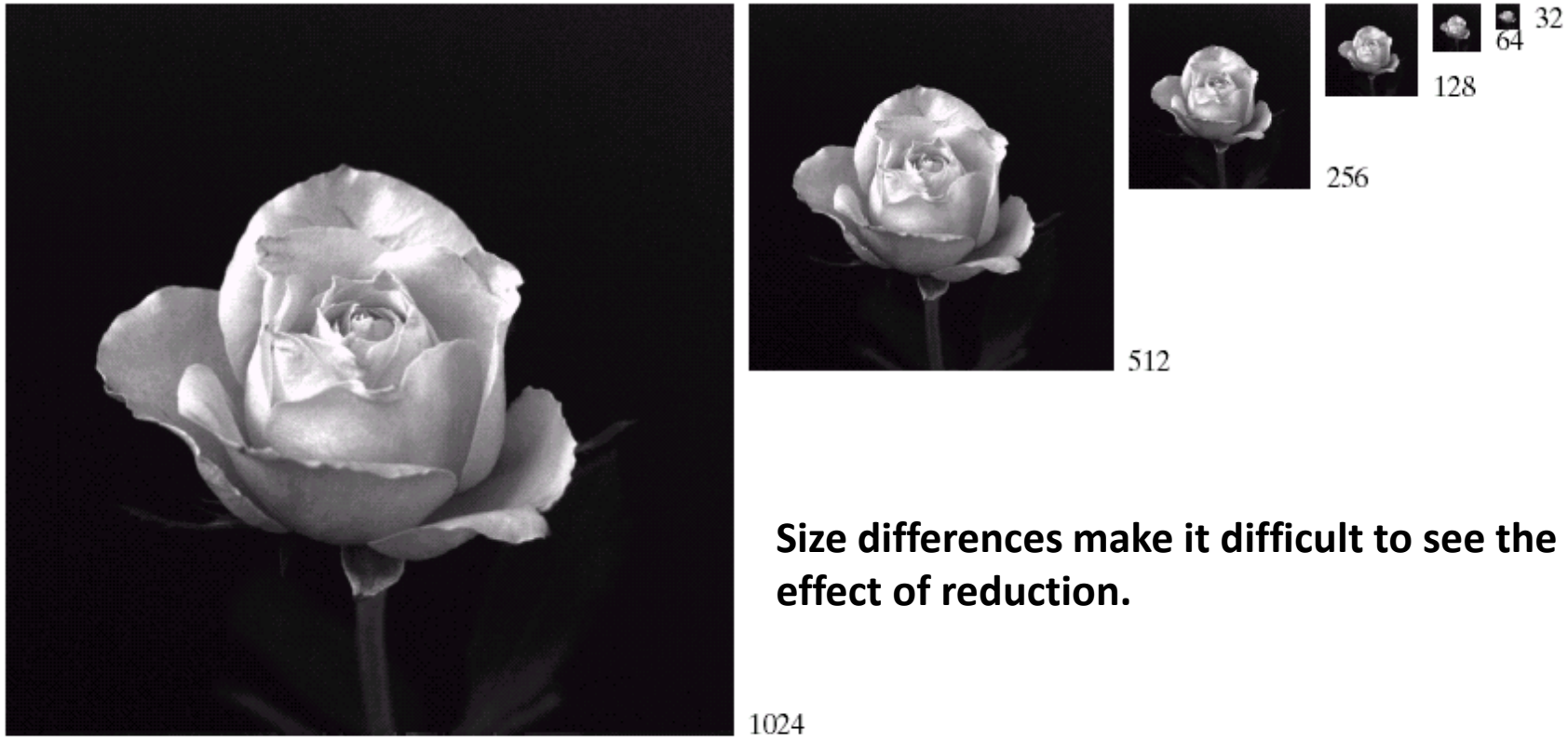
- How many bits to represent each pixel.
  - 8 bit
  - 12 bits
  - 32 bits
  - Etc.

Second Measure: Image resolution

- This affects the size of the image
- Number or rows and columns in the image

Same Pixel Size (gray levels), different Sizes

Subsampling: Size



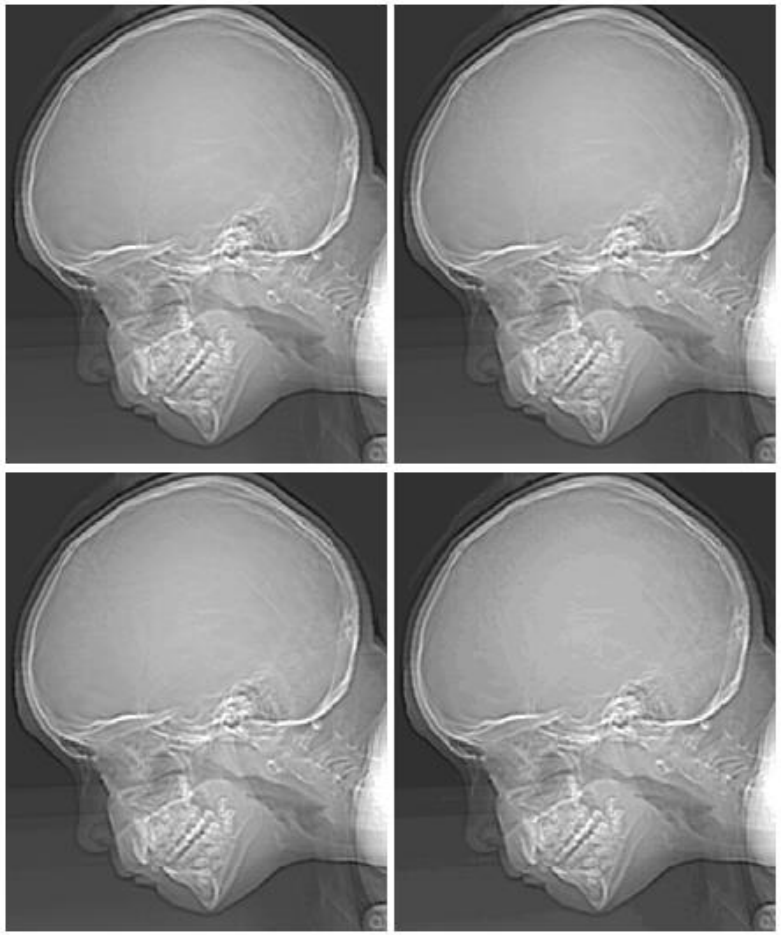
Size differences make it difficult to see the effect of reduction.

FIGURE 2.19 A 1024 × 1024, 8-bit image subsampled down to size 32 × 32 pixels. The number of allowable gray levels was kept at 256.

Matlab Exercise

- Use Matlab to change the size of an image
  - help imresize

Varying Gray Level Resolution

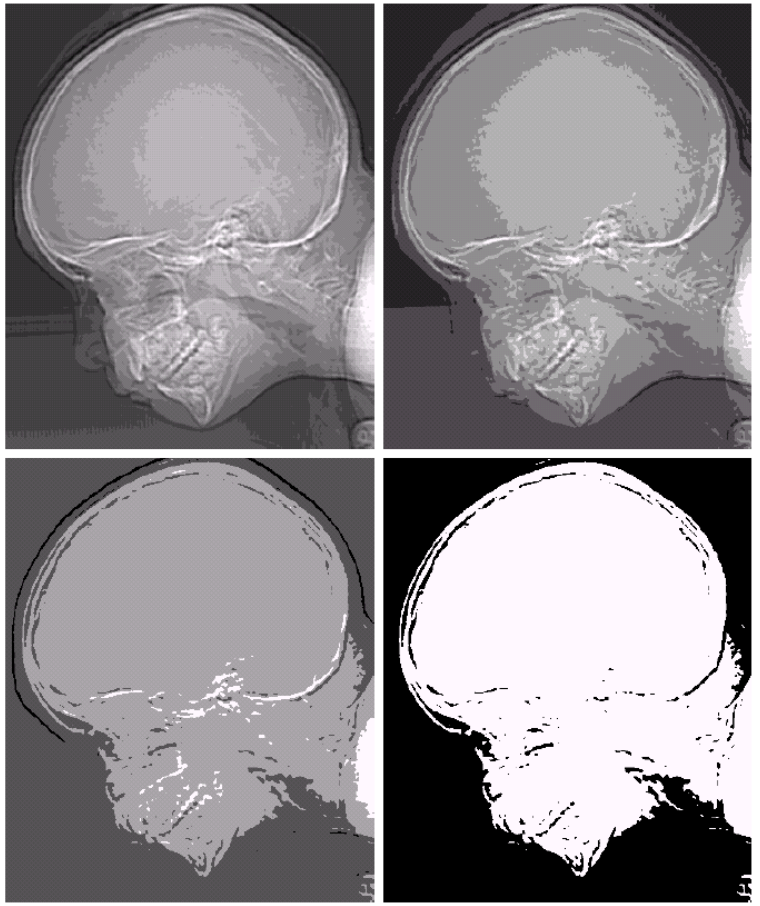


a b
c d

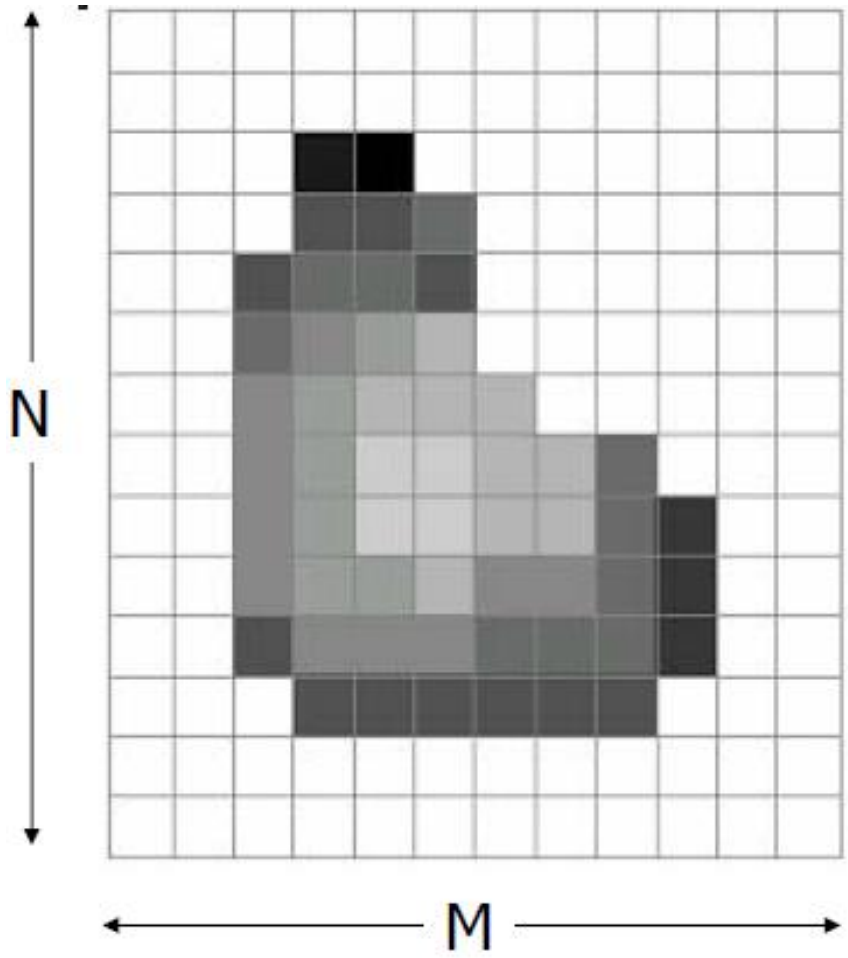
FIGURE 2.21
(a) 452 × 374, 256-level image. (b)–(d) Image displayed in 128, 64, and 32 gray levels, while keeping the spatial resolution constant.

e f
g h

FIGURE 2.21
(Continued)
(e)–(h) Image displayed in 16, 8, 4, and 2 gray levels. (Original courtesy of Dr. David R. Pickens, Department of Radiology & Radiological Sciences, Vanderbilt University Medical Center.)



Number of bits



- The number of gray levels typically is an integer power of 2

$$L = 2^k$$

- Number of bits required to store a digitized image

$$b = M \times N \times k$$

$$= N^2 \times k \text{ (when } M=N)$$

Resolution

- Resolution (how much you can see the detail of the image) depends on sampling and gray levels.
- The bigger the sampling rate (***n***) and the gray scale (***g***), the better the approximation of the digitized image from the original.
- The more the quantization scale becomes, the bigger the size of the digitized image.

Questions